EXPERIMENT NO. 10:

Write a Python Program to Implement Expectation & Maximization (EM) Algorithm

Aim:

To implement the Expectation and Maximization (EM) Algorithm in Python for clustering data using a Gaussian Mixture Model (GMM).

Procedure:

1. **Import the required Python libraries.**
2. **Create a sample dataset.**
3. **Initialize the Gaussian Mixture Model (GMM).**
4. **Train the model using the Expectation-Maximization (EM) algorithm.**
5. **Predict the cluster labels for the data.**
6. **Display the cluster assignments.**
7. **Display the cluster means obtained after training.**

Program:

```
# Expectation Maximization (EM) Algorithm


import numpy as np

from sklearn.mixture import GaussianMixture


# Sample Dataset

X = np.array([

   [1, 2], [2, 1], [1, 1], [2, 2],

   [8, 8], [9, 8], [8, 9], [9, 9]

])


# Create Gaussian Mixture Model

gmm = GaussianMixture(n_components=2, random_state=42)


# Train the model

gmm.fit(X)
```

```
# Predict cluster labels

labels = gmm.predict(X)


print("Cluster Labels:")

print(labels)


print("\nCluster Means:")

print(gmm.means_)
```

Output:

```
••• Cluster Labels:
    [0 0 0 0 1 1 1 1]

    Cluster Means:
    [[1.5 1.5]
     [8.5 8.5]]
```

Result:

The Expectation-Maximization (EM) Algorithm was successfully implemented in Python using the Gaussian Mixture Model (GMM). The algorithm correctly clustered the given dataset and identified the corresponding cluster means.